Gray-Zone Maritime Control in the First Island Chain: A Networked Escalation Game

Workshop 1 Report (Deliverable 2)

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1 Setting and research questions

Contested system: China and the United States compete for day-to-day control of the waters inside the First Island Chain (Senkakus, Taiwan Strait, Luzon Strait, Scarborough and Second Thomas Shoals, Paracels) *below the threshold of war*. China claims most of the South China Sea inside its nine-dash line, contests the Senkakus, and presses these claims through persistent presence with coast guard and maritime militia (CCG) hulls backed by stronger, more heavily equipped navy (PLAN) ships. The U.S. rejects maritime claims beyond what international law allows, including territorial seas around artificial islands, and asserts freedom of the high seas, open shipping lanes, and area access through a smaller number of destroyers (DDG) and littoral combat ships (LCS) on presence and freedom-of-navigation patrols (Helmus et al., 2024; Green et al., 2017). Each side decides, every two days, *where* to station its ships on a network of zones and *how* to behave there: patrol (P) or assert (A: shadowing; close maneuvering; querying and warning; and, for the U.S. only, freedom-of-navigation operations).

Why strategic interaction is central: Control of a zone is a function of the ratio between local forces. Therefore, each actor's best allocation depends on the other's allocation. Escalation risk is shared between both parties, where incidents need both sides present and escalation to war costs both sides. Neither side sees the other's entire fleet disposition: each reliably sees only the forces present where it also maintains forces.

Research questions:

RQ1. *Control vs. escalation.* Each side's allocation is a MIP with an escalation-risk budget ε_i. How much control does a larger ε_i buy, at what war risk, and which budgets are equilibrium choices when both sides pick them?

RQ2. *Strategic vs. heuristic play.* Do game-theoretic policies (equilibrium and maximin strategies, exact best responses) beat random and rule-based ones, and can mutual restraint be sustained in repeated play?

RQ3. *Robustness and redesign.* Is the best policy or Blue design under nominal conditions also the most robust to weather (Nature's scenario ω), in expected and worst-case terms?

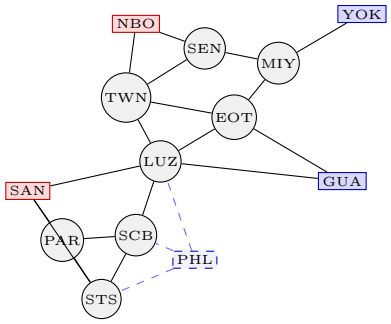
Outcomes (averaged over 200 seeds): Blue's value-weighted control share $\bar{c}_{\mathsf{B}} = \frac{1}{\tau}\sum_t \sum_v V_v^{\mathsf{B}} c_v^t / \sum_v V_v^{\mathsf{B}} \in [-1, 1]$, war rate, payoffs $u_{\mathsf{B}}, u_{\mathsf{R}}$, and for RQ3 the expected and worst-case payoff over ω.

Changes from the proposal and how we used the feedback:

Feedback	What we changed (section)
Model too high level; define it mathematically	Full game on a graph: state, actions, information, dynamics, payoffs (§3)
Q1 is a feasible region; can be a MIP	Best response solved exactly as a MIP; the old aggressive/moderate/conservative types are values of its risk budget ε_i (§3.6)
Research questions more ambitious	RQ1: frontier and equilibrium of a budget game; RQ2: equilibrium play; RQ3: robustness and redesign
Define the resilient redesign	Three Blue designs (§3.8), tested in §4.5
Propose values for missing data	All values in Tables 1 and 2, marked as assumptions
Show preliminary results	Analytical results and 77 simulated configurations × 200 seeds (§4)
Prof. Sapsis: add external uncertainty (weather) and test robustness	Sea state is Nature's move; four weather scenarios; expected and worst-case payoff (§3.2, §4.5)

2 Literature and current practice

Documented practice: Helmus et al. (2024) catalog China's gray-zone tactics: persistent CCG and militia presence, shadowing, blocking resupply at Second Thomas Shoal, water cannon, and non-navy hulls that stay below the threshold of armed response. Green et al. (2017) argue that deterrence below war depends on presence and on managing escalation. U.S. practice is routine presence and freedom-of-navigation patrols, with encounters managed by the Code for Unplanned Encounters at Sea (WPNS, 2014). We take from these the action set (presence, patrol vs. assert), the asymmetric hull types, and the idea that incidents, not shots, drive escalation. **Models:** Wasser et al.



One period t (timeline):

1. Nature draws the sea state w_r^t of each region (scenario ω).
2. Each player observes its own ships, ℓ^t, c^t, w^t, and detects opponent ships with probability $q_{i,v}$.
3. Ships whose endurance is nearly spent are forced toward home.
4. Both players choose moves and postures *simultaneously*.
5. Moves resolve under fuel budgets; endurance updates.
6. Control indices update; incidents are drawn; ℓ moves up or down.
7. Stage payoffs are paid; the game stops if $\ell = L$ (war) or $t = T = 60$.

Figure 1: Left: theater graph G (schematic). Squares: home ports (blue H_{B}, red H_{R}); circles: contested zones. Dashed: Subic/EDCA port and edges added by redesign D1. Right: sequence of play.

(2019) design a seminar gray-zone game with explicit objectives and action menus; we make every adjudication rule computable. The U.S. Coast Guard schedules randomized patrols from a game model (Shieh et al., 2012); zero-sum network games model transit and blockade (Washburn and Wood, 1995); escalation as a shared, manipulated risk (Schelling, 1966) is what our incident hazard makes explicit. **What we add:** a general-sum game where both sides contest control at once and war is a joint cost, with explicit rules so strategies and designs can be compared over thousands of runs. Numerical values are our assumptions (Table 1), not documented data.

3 Game formulation

This section defines the game in the order a reader needs it: who plays and on what map (§3.1), what describes the situation and what each side can see (§3.2), what each side decides (§3.3), how the situation then changes (§3.4), how each side scores (§3.5), and how a side chooses its moves (§3.6–3.7). Table 1 lists the main symbols with their meanings and values, and Table 2 the zone values and weather scenarios. All numbers are our assumptions; there is no public data for them.

3.1 Players, map and time

Players: $\mathcal{I} = \{\mathsf{B}, \mathsf{R}\}$. Blue is the U.S. Navy. Red is China, with navy (PLAN) ships and coast guard or maritime militia hulls, grouped as CCG. Ship types are $\mathcal{K}_{\mathsf{B}} = \{\text{DDG}\}$ and $\mathcal{K}_{\mathsf{R}} = \{\text{PLAN}, \text{CCG}\}$, with fleets of 10 DDG, 8 PLAN and 6 CCG.

Map: the theater is an undirected graph $G = (\mathcal{V}, \mathcal{E})$ (Fig. 1, Lecture 2). Its nodes are home ports, where ships refuel, and eight contested zones $\mathcal{C} = \{\text{SEN}, \text{MIY}, \text{TWN}, \text{EOT}, \text{LUZ}, \text{SCB}, \text{STS}, \text{PAR}\}$ (Senkakus, Miyako Strait, Taiwan Strait, east of Taiwan, Luzon Strait, Scarborough Shoal, Second Thomas Shoal, Paracels). Blue's home ports are $H_{\mathsf{B}} = \{\text{YOK}, \text{GUA}\}$ (Yokosuka, Guam) and Red's are $H_{\mathsf{R}} = \{\text{NBO}, \text{SAN}\}$ (Ningbo, Sanya); neither side may enter the other's ports. An edge joins two nodes that a ship can sail between in one period. The zones form three regions that share weather: North $= \{\text{SEN}, \text{MIY}\}$, Center $= \{\text{TWN}, \text{EOT}, \text{LUZ}\}$ and South $= \{\text{SCB}, \text{STS}, \text{PAR}\}$. Zone v is worth V_v^i to player i (Table 2); the two sides value zones differently (Scarborough Shoal and the Paracels matter more to Red).

Time: the game is played in periods $t = 1, \ldots, T$ of two days each, with $T = 60$ (about four months). Payoffs in later periods are discounted by $\delta = 0.98$ per period (Lectures 5–6). All ships start in port with full endurance: Blue's DDGs are split 5/5 between YOK and GUA, and each Red type is split evenly between NBO and SAN.

3.2 State, information and weather

State: the situation at the start of period t is $s^t = (x^t, d^t, c^t, \ell^t, w^t)$:

- $x_{k,v}^t \in \mathbb{Z}_{\geq 0}$: the number of type-$k$ ships at node v.
- d^t: each ship's *endurance*, the number of periods it can still stay at sea (at most D_k: 10 for a DDG, 8 for a PLAN ship, 6 for a CCG hull). It drops by one each period at sea and refills after a full period in port.
- $c_v^t \in [-1, 1]$: the *control index* of zone v. $+1$ means Blue is fully in charge there, -1 means Red is, and 0 means neither is. It changes gradually as presence builds or fades (§3.4).
- $\ell^t \in \{0, 1, \ldots, L\}$: the *escalation level*, a ladder of tension from calm (0) to armed conflict ($L = 5$), which ends the game. The game starts at $\ell^1 = 1$.
- $w_r^t \in \{\text{calm}, \text{rough}, \text{storm}\}$: the sea state in region r.

Information: each player sees its own ships and the public quantities c^t, ℓ^t and w^t, but not where the opponent's

State, beliefs and choices	
$x^t_{k,v}$	type-k ships at node v
d	periods a ship can still stay at sea
c^t_v	control of v: $+1$ Blue, -1 Red
ℓ^t	escalation level (war at L)
w^t_r	sea state of region r
$\hat{x}^t_{-i,v}$	i's last-seen opponent count at v
$p_{i,v}$	posture: patrol (P) or assert (A)
$f^t_{k,uv}$; m^t_i	ships sailing $u \to v$; moves made
$S_{i,v}$; ϕ_v	local strength; local balance
h_v; ρ_v	incident hazard; probability
g^t_i; u_i	stage payoff; total payoff
$y_{s,w}$	MIP: ship s goes to node w
$b_{v,(n,p)}$	MIP: n hulls, posture p at v
ω; π_ω	weather scenario; its weight
V^i_v; σ_i	value of v to i; $+1$ Blue, -1 Red

Parameters (our assumptions)		
T; δ	periods (2 days each); discount	60; 0.98
N_k	fleet size	10 DDG; 8 PLAN, 6 CCG
D_k	endurance when refueled (periods)	10 DDG; 8 PLAN; 6 CCG
F_i	ship moves per period	8 Blue; 10 Red
w_k	strength per hull (CCG in rough seas)	1 DDG, PLAN; 0.6 CCG (0.3)
α	strength multiplier when asserting	1.5
κ; λ	presence damping; control rate	1; 0.3
h_0	baseline incident hazard	0.006
$h^{\mathsf{B}}_{\mathrm{A}}$; $h^{\mathsf{R}}_{\mathrm{A}}$	extra hazard if asserting	0.03; 0.036 PLAN, 0.012 CCG
L; ℓ^1	war level; starting level	5; 1
μ	chance a quiet period lowers ℓ	0.25 (D3: 0.5 at $\ell \geq 3$)
γ; η	move cost; tension cost per level	0.1; 1
W	war penalty (both sides)	200
$q_{i,v}$	detection: here / nearby / far	0.95 / 0.6 / 0.2, $\times$0.7 rough
Λ; β	MIP: incident price; persistence	20; 0.96
ε_i	escalation-risk budget	0.05 / 0.15 / 0.40
ζ_D	Blue's redesign cost per period	D1 0.3; D2 0.5; D3 0

Table 1: Notation. Left: quantities that describe the game as it is played. Right: parameter values, all of them our assumptions (no public data); hazards and μ were set so that war is uncommon but not rare (0–34% of games across the configurations tested).

Zone	TWN	SEN	MIY	EOT	LUZ	SCB	STS	PAR
V^{B}_v	10	6	4	4	6	5	5	2
V^{R}_v	10	6	3	4	5	7	6	6
c^1_v	0	0.2	0.3	0.2	0.1	-0.2	0	-0.4

ω	calm	nominal	typhoon	extreme
π_ω	0.25	0.40	0.25	0.10
$p_s(\omega)$	0	0.02	0.06	0.12
$p_r(\omega)$	0	0.20	0.35	0.50

Table 2: Left: zone values V^i_v and starting control c^1_v. Right: weather scenarios: weight π_ω, chance per period that a storm starts in a region $p_s(\omega)$, and chance of rough seas $p_r(\omega)$.

ships are now. Each period it detects the opponent's ships at node v with probability

$$q_{i,v} = \begin{cases} 0.95 & \text{if } i \text{ has a ship at } v, \\ 0.6 & \text{if } i \text{ has a ship at a neighboring node, or } v \text{ borders one of } i\text{'s home ports,} \\ 0.2 & \text{otherwise (satellite),} \end{cases}$$

multiplied by 0.7 in rough or stormy seas. A detection shows the opponent's ship count of each type at v and its posture there. Player i's *belief* $\hat{x}^t_{-i,v}$ is the last count it detected at v (exact at the start, when every ship is in port). So both sides choose their moves at information sets that do not reveal where the opponent is now (Lectures 5–8).

Weather (Nature's move): the weather is a random factor that neither side controls. We study four weather scenarios $\omega \in \Omega = \{\text{calm, nominal, typhoon, extreme}\}$ with weights $\pi_\omega = (0.25, 0.40, 0.25, 0.10)$ (Lecture 1). In scenario ω, each period and in each region not already in a storm, a storm starts with probability $p_s(\omega)$; otherwise the sea is rough with probability $p_r(\omega)$ and calm if not (Table 2). A storm lasts three periods. It closes the region's zones (no ship may enter, and ships inside stay put), stops incidents there, and freezes their control. Rough seas halve the strength of CCG hulls (§3.4) and degrade detection. §4.5 scores each policy and design by its expected payoff over ω and by its worst case (Lecture 1).

3.3 Actions and constraints

Actions: both players choose at the same time. Each period player i chooses

- a destination for each of its ships: stay, or sail one edge to a neighboring node; and
- a posture $p_{i,v} \in \{\mathrm{P}, \mathrm{A}\}$ in each zone v where it will have ships. *Patrol* (P) is routine presence. *Assert* (A) stands for the actions listed in §1 (shadowing, close maneuvering, querying and warning, and Blue's freedom-of-navigation operations), treated as one posture. Asserting makes the side's ships in that zone count 1.5 times as much for control, but makes incidents more likely (§3.4).

Constraints: write $f^t_{k,uv}$ for the number of type-k ships that sail from u to v in period t, and $\mathcal{N}(u)$ for the neighbors of node u. Then

$$x^{t+1}_{k,u} = x^t_{k,u} + \textstyle\sum_{v \in \mathcal{N}(u)} f^t_{k,vu} - \sum_{v \in \mathcal{N}(u)} f^t_{k,uv} \qquad \text{(flow conservation: ships are moved, never created; Lecture 2)}$$

$$\textstyle\sum_{k \in \mathcal{K}_i} \sum_{(u,v)} f^t_{k,uv} \leq F_i \qquad \text{(fuel: at most } F_{\mathsf{B}} = 8 \text{ and } F_{\mathsf{R}} = 10 \text{ ship moves per period)}$$

and three more rules:

- *Endurance.* A ship may not sail to a node from which it could not get home on its remaining endurance, unless the move brings it closer to home. A ship whose endurance has fallen to its distance from home is on *forced return*: it sails one edge toward port each period, then spends one full period in port to refit, after which d resets to D_k. Forced moves use fuel first; the player decides only for its other (*free*) ships.
- *Ports and storms.* No ship may enter the opponent's home ports or a storm-closed zone.
- *Infeasible orders.* A planned move that breaks a rule, or that exceeds the fuel left, is not carried out: the ship stays where it is.

3.4 How the state changes: control, incidents and escalation

After the moves, steps 1–3 happen in every contested zone v that is not in a storm, and step 4 updates the escalation level once for the whole theater.

1. Local strength: each side's weight of presence in the zone is

$$S_{i,v} = \alpha^{\mathbf{1}[p_{i,v}=\mathrm{A}]} \sum_{k\in\mathcal{K}_i} w_k\, x_{k,v},$$

a weighted count of its hulls there. A DDG or a PLAN ship counts $w_k = 1$; a CCG hull counts 0.6 (it is smaller and lightly armed), and only 0.3 in rough seas. Asserting multiplies the side's strength by $\alpha = 1.5$.

2. Control: the *local balance* $\phi_v = (S_{\mathsf{B},v} - S_{\mathsf{R},v})/(S_{\mathsf{B},v} + S_{\mathsf{R},v} + \kappa)$ lies between -1 and 1 and is positive when Blue is locally stronger. The constant $\kappa = 1$ keeps a small presence from counting as full control: one DDG alone gives $\phi_v = 0.5$. Each period, control moves a fraction $\lambda = 0.3$ of the way toward the local balance:

$$c_v^{t+1} = (1-\lambda)\, c_v^t + \lambda\, \phi_v. \tag{1}$$

So control is earned by staying present for several periods and fades when a side leaves. In a storm, c_v does not change.

3. Incidents: an incident (a collision, an unsafe intercept, a standoff that goes wrong) can happen only where both sides have ships. It occurs with probability $\rho_v = 1 - e^{-h_v}$, where the *hazard* h_v is

$$h_v = \Big(h_0 + h_{\mathrm{A}}^{\mathsf{B}}\,\mathbf{1}[p_{\mathsf{B},v} = \mathrm{A}] + h_{\mathrm{A},v}^{\mathsf{R}}\,\mathbf{1}[p_{\mathsf{R},v} = \mathrm{A}]\Big) \cdot \min\{S_{\mathsf{B},v}, S_{\mathsf{R},v}\} \cdot \big(1 + \ell^t/2\big). \tag{2}$$

The three factors read as follows. (i) Routine encounters carry a small baseline hazard $h_0 = 0.006$, and each side that asserts adds to it: $h_{\mathrm{A}}^{\mathsf{B}} = 0.03$ for Blue; for Red, $h_{\mathrm{A},v}^{\mathsf{R}} = 0.036\,\theta_v + 0.012\,(1-\theta_v)$, where θ_v is the PLAN share of Red's strength at v, so pressure from coast guard hulls is less escalatory than pressure from navy ships. (ii) More ships in contact mean more encounters, and the smaller of the two forces limits how many there can be. (iii) Incidents are more likely when tension is already high: at $\ell = 3$ the hazard is 2.5 times its value at $\ell = 0$.

4. Escalation: each incident raises ℓ by one. A period with no incident anywhere lowers ℓ by one with probability $\mu = 0.25$. If ℓ reaches $L = 5$, the crisis has become an armed conflict and the game ends.

3.5 Payoffs

Stage payoff: at the end of period t, player i receives

$$g_i^t = \underbrace{\sigma_i \textstyle\sum_{v\in\mathcal{C}} V_v^i\, c_v^{t+1}}_{\text{control}} - \underbrace{\gamma\, m_i^t}_{\text{fuel}} - \underbrace{\eta\, \ell^{t+1}}_{\text{tension}} - \underbrace{\zeta_D\, \mathbf{1}[i = \mathsf{B}]}_{\text{redesign cost}}. \tag{3}$$

- *Control.* $\sigma_{\mathsf{B}} = +1$ and $\sigma_{\mathsf{R}} = -1$ turn the shared control index into each side's gain: Blue gains V_v^{B} for each unit of c_v above zero, and Red gains V_v^{R} for each unit below zero.
- *Fuel.* m_i^t is the number of ship moves player i made in period t, at $\gamma = 0.1$ each.
- *Tension.* $\eta = 1$ per escalation level per period stands for the diplomatic and economic cost of a crisis.
- *Redesign cost.* ζ_D is the running cost of Blue's redesign D (§3.8).

Total payoff: the game ends at τ, the first period with $\ell^\tau = L$ (war), or at T. Player i's payoff is the discounted sum of its stage payoffs, minus a war penalty $W = 200$ if the game ends in war:

$$u_i = \sum_{t=1}^{\tau} \delta^{t-1} g_i^t - W\,\mathbf{1}[\ell^\tau = L]. \tag{4}$$

Both sides pay W, so the game is general-sum, not zero-sum (Lecture 3): both prefer to avoid war. For scale, Blue earns at most $\sum_v V_v^{\mathsf{B}} = 42$ in a period in which it fully controls every zone, so a war costs about five such periods.

3.6 How a player decides: the best-response MIP with a risk budget

Our main decision rule, *MIP-BR*, chooses each period the moves and postures that score best given what the player believes, while capping its risk of incidents. It is a best response (Lectures 1 and 3) to the opponent's believed positions, written as a mixed-integer program (Lecture 2) and solved with Gurobi, as in the lecture code.

What the player takes as given: its beliefs $\hat{x}_{-i}$, the opponent postures it last saw (patrol if never seen), the sea state, the escalation level ℓ^t, and its own ships. Ships on forced return are fixed; $\bar{x}_{k,v}$ counts them by where they will be.

Decision variables: $y_{s,w} \in \{0,1\}$ equals 1 if free ship s goes to node w (it may stay, sail home, or sail to a neighbor from which it can still get home). Control and incident risk are nonlinear in the ship counts (a ratio and an exponential), so we list every count a zone could end up with: $b_{v,(n,p)} \in \{0,1\}$ equals 1 if the player will have $n = (n_k)_{k \in \mathcal{K}_i}$ hulls in zone v with posture p. Exactly one option is chosen per zone, and each option's value and risk are computed in advance, which keeps the program linear.

Value of an option:

$$a_v(n,p) = V_v^i \, \beta \, \sigma_i \phi_v(n,p) - \Lambda \, \rho_v(n,p), \qquad \beta = \frac{\lambda}{1 - \delta(1-\lambda)} \approx 0.96.$$

Here $\phi_v(n,p)$ and $\rho_v(n,p)$ are the local balance and the incident probability of §3.4 if the player brings n hulls with posture p and the opponent is where the player believes it is. The factor β turns one period of local balance into its total discounted effect on control: by (1), a balance ϕ_v moves c_v by $\lambda\phi_v$ now, a fraction $1-\lambda$ of that change survives each later period, and the discounted sum is $\lambda \sum_{j \geq 0} (\delta(1-\lambda))^j = \beta$. $\Lambda = 20$ is the price the player puts on one expected incident, since an incident raises ℓ (costing η every period) and brings war closer.

The program:

$$\begin{aligned}
\max_{y,\, b} \quad & \textstyle\sum_{v \in \mathcal{C}} \sum_{(n,p)} a_v(n,p)\, b_{v,(n,p)} - \gamma \sum_s \sum_{w \neq \mathrm{loc}(s)} y_{s,w} && \text{(control value − incident cost − fuel cost)} \\
\text{s.t.} \quad & \textstyle\sum_w y_{s,w} = 1 \quad \forall s && \text{(each free ship gets one destination)} \\
& \textstyle\sum_s \sum_{w \neq \mathrm{loc}(s)} y_{s,w} \leq F_i - \text{(forced moves)} && \text{(fuel)} \\
& \textstyle\sum_{(n,p)} b_{v,(n,p)} = 1 \quad \forall v && \text{(one option per zone)} \\
& \textstyle\sum_{(n,p)} n_k \, b_{v,(n,p)} = \bar{x}_{k,v} + \sum_{s \text{ of type } k} y_{s,v} \quad \forall v, k && \text{(the option matches the moves)} \\
& \textstyle\sum_{v \in \mathcal{C}} \sum_{(n,p)} \rho_v(n,p)\, b_{v,(n,p)} \leq \varepsilon_i \, (1 - \ell^t / L) && \text{(escalation-risk budget)} \\
& y,\, b \text{ binary.}
\end{aligned}$$

The escalation-risk budget: the left side of the last constraint is the player's expected number of incidents this period. It accepts at most ε_i of them, and the cap shrinks to zero as ℓ approaches war. The budget defines the feasible region, the set of allocations the player considers safe enough. The proposal's conservative, moderate and aggressive commanders become budgets $\varepsilon_i \in \{0.05, 0.15, 0.40\}$, and RQ1 varies ε_i. If forced returns alone break the budget, a slack variable with a large penalty keeps the program feasible. The program looks only one period ahead; β is how it values the future effect of control.

3.7 The template game and the policies we compare

Template game (Lectures 3–4): the full game has far too many strategies to solve for an equilibrium directly, so we also study a small strategic-form version. Each player picks one of 14 *templates*: one of seven ways to split its deployable ships across North/Center/South (60/20/20, 20/60/20, 20/20/60, 40/40/20, 40/20/40, 20/40/40 or 34/33/33 percent), combined with one posture (P or A) in all its zones. Deployable means 70% of each fleet, spread evenly over the zones of a region (Red uses PLAN and CCG in proportion; fractional hulls are allowed). If both sides kept their templates, control would settle at $c_v = \phi_v$ (set $c_v^{t+1} = c_v^t$ in (1)). So the payoff of a template pair $(a_{\mathsf{B}}, a_{\mathsf{R}})$ is that settled control minus the incident cost, in calm seas at $\ell = 1$:

$$M_i(a_{\mathsf{B}}, a_{\mathsf{R}}) = \sigma_i \textstyle\sum_{v \in \mathcal{C}} V_v^i \, \phi_v - \Lambda \sum_{v \in \mathcal{C}} \rho_v.$$

This gives a finite game $\Gamma = \langle \mathcal{I}, (S_i), (M_i) \rangle$ with $|S_i| = 14$, which we analyze with iterated elimination of strictly dominated strategies (IESDS), enumeration of pure Nash equilibria, the maximin LP and the correlated-equilibrium LP.

Policies: in the simulations (§4) each side follows one of these rules:

- *random*: random legal moves and postures (a benchmark).

- *rule-based* ("moderate"): sends ships to the zones with the best value per unit of opponent strength; asserts only where it believes it outnumbers the opponent and $\ell \leq 2$; then drops assertions and pulls ships back until its expected incidents fit the risk budget $\varepsilon = 0.15\,(1 - \ell/L)$.
- *greedy*: a fast heuristic for the MIP of §3.6 (same objective and budget, no solver).
- *MIP-BR*(ε): solves the MIP exactly each period.
- *equilibrium*: plays its part of a pure Nash equilibrium of Γ, steering its free ships toward the template's split (more ships to the more valuable zones of a region). Γ has two pure equilibria; we use the one with the higher total payoff, which is an assumption.
- *maximin*: every three periods, draws a template from its maximin mixed strategy of Γ, the one that guarantees the best payoff whatever template the opponent plays.

The two template policies switch to patrol when $\ell > 3$ and drop assertions that break their risk budget.

3.8 Resilient redesigns (Blue)

A redesign changes Blue's system, not the way Blue plays it. In the terms of Lecture 1, D1 is a structural change (an alternative path) and D2 and D3 are operational ones (monitoring and rapid response). Each is compared with the baseline D0 in every weather scenario:

- **D1 dispersed basing:** a third Blue port at Subic Bay (EDCA access), linked to SCB, LUZ and STS, which gives alternative paths and shorter trips to the South. Blue starts with 4/3/3 DDGs at YOK/GUA/Subic. Cost $\zeta_{\mathrm{D1}} = 0.3$ per period.
- **D2 redundant sensing:** Blue detects in every zone with probability at least 0.6, also in bad weather. Cost $\zeta_{\mathrm{D2}} = 0.5$ per period.
- **D3 crisis hotline** (CUES-style communication): once $\ell \geq 3$, a quiet period lowers ℓ with probability $\mu = 0.5$ instead of 0.25, and assert hazards are halved. We charge no cost yet ($\zeta_{\mathrm{D3}} = 0$), and it helps Red as much as Blue.

Simplifying assumptions: two unitary players; control is a force-ratio index; incidents are the only path to war; beliefs are last-seen counts; best responses look one period ahead.

4 Preliminary results

§4.1–4.2 are analytical (template game); the rest use 200 seeds per configuration, the same for every policy. Unless stated, Red plays MIP-BR with $\varepsilon_{\mathsf{R}} = 0.15$ in nominal weather.

4.1 Small example: patrol or assert, and sustainable restraint

Fix the force splits that survive dominance (§4.2): Blue center-heavy, Red south-heavy. Each side then only chooses P or A. The resulting 2×2 sub-matrix of Γ, normalized so (P,P) = $(0, 0)$, has (P,A) = $(-9.61,\ 1.20)$, (A,P) = $(0.28,\ -10.45)$ and (A,A) = $(-11.41,\ -11.33)$ (Blue, Red). Asserting against a patroller pays a little (one-period gain $G_{\mathsf{B}} = 0.28$, $G_{\mathsf{R}} = 1.20$), but mutual assertion costs more than patrolling alone, so $BR_i(\mathrm{P}) = \mathrm{A}$ and $BR_i(\mathrm{A}) = \mathrm{P}$: the same structure as the shared-compute example (Lecture 3). There are two pure NE, (A,P) and (P,A), and a mixed NE (Blue asserts w.p. 0.58, Red 0.13) with mutual assertion w.p. 0.08 and payoffs $v_{\mathsf{B}} = -1.25$, $v_{\mathsf{R}} = -6.06$. The welfare-maximizing correlated equilibrium (CE LP) puts 0.40 on (P,P), 0.54 on (A,P), 0.06 on (P,A) and *zero* on (A,A): a mediator (e.g. a hotline) removes mutual assertion.

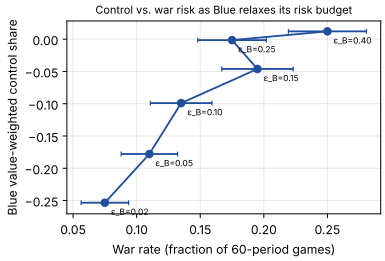
In the repeated game (Lectures 5–6), consider grim trigger: both patrol; after any assertion, both play the mixed NE forever. By the one-stage deviation principle, patrolling is optimal iff $(1 - \delta)G_i + \delta v_i \leq 0$, i.e. $\delta \geq G_i/(G_i - v_i)$, which is 0.18 for Blue and 0.17 for Red. With $\delta = 0.98$, **mutual restraint is a subgame perfect equilibrium**, although our one-period-ahead policies cannot see this yet.

4.2 Template game: dominance, equilibria and the effect of the escalation cost

IESDS reduces Γ from 14×14 to 2×4: Blue keeps only the center-heavy split (20% North, 60% Center, 20% South), Red the south- or center/south-heavy splits. There are two pure NE: (Blue C-heavy assert, Red S-heavy patrol), payoffs $(0.39, -9.27)$, and (Blue C-heavy patrol, Red C/S assert), $(-14.44, 2.71)$; the *equilibrium* policy plays the first (higher total payoff; this selection is an assumption). Without the escalation cost, the zero-sum maximin LP has a pure equilibrium where *both* sides concentrate on the Center and assert (value -2.26): the shared incident cost is what makes at most one side assert.

4.3 RQ1: control versus war risk, and the risk-budget game

Frontier: Raising ε_{B} from 0.02 to 0.15 lifts Blue's control share from -0.25 to -0.05 while the war rate rises from 0.07 to 0.20; going to 0.40 adds only 0.06 of control (Fig. 2). Most of what a looser budget buys is bought by



B\R	cons. 0.05	mod. 0.15	aggr. 0.40
cons. 0.05	−194, 199 war 0.10	−303, 315 0.11	−308, 318 0.12
mod. 0.15	18, −36 war 0.15	−144, 101 0.20	−192, 140 0.21
aggr. 0.40	74, −80 war 0.14	−83, 16 0.25	**−151, 39** **0.34**

Figure 2: Left: Blue's control share vs. war rate as Blue's risk budget ε_B grows (Red fixed at 0.15). Right: the risk-budget game, (u_B, u_R) and war rate when both sides choose ε (s.e. $\approx$ 5–12). Bold: the unique Nash equilibrium.

$\varepsilon \approx 0.15$–0.25.

Risk-budget game: When both sides pick a budget, "aggressive" is a best response to every opponent choice for both players (Red vs. conservative Blue, 318 vs. 315, is within noise), so (aggressive, aggressive) is the unique pure NE. It has the *highest* war rate (0.34 vs. 0.10 when both are conservative) and total payoff −112 vs. +12 at (conservative, moderate): a prisoner's dilemma (Lecture 3). So restraint must come from repetition or coordination (§4.1).

4.4 RQ2: strategic versus heuristic play

Blue \ Red	rule-based (moderate)	MIP-BR	equilibrium	worst case over Red
random	−283 / −0.14 / 0.15	−319 / −0.16 / 0.33	−164 / −0.08 / 0.06	−319
rule-based (moderate)	−122 / −0.05 / 0.04	−184 / −0.09 / 0.06	35 / +0.06 / 0.04	−184
greedy	−70 / +0.01 / 0.12	−137 / −0.05 / 0.14	65 / +0.08 / 0.03	−137
MIP-BR	−75 / +0.01 / 0.13	−144 / −0.05 / 0.20	53 / +0.08 / 0.08	−144
equilibrium	−72 / −0.01 / 0.07	**−116** / −0.03 / 0.10	21 / +0.05 / 0.06	**−116**
maximin	−170 / −0.09 / 0.04	−221 / −0.12 / 0.08	−35 / −0.01 / 0.00	−221

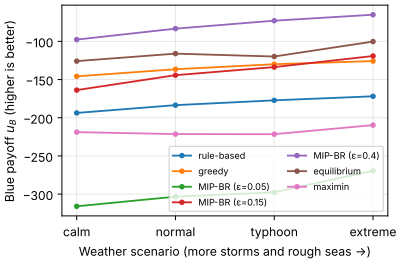
Table 3: Blue payoff u_B / control share / war rate (s.e. of u_B is 1–7). Nominal weather, design D0.

Greedy, MIP-BR and equilibrium beat the random and rule-based benchmarks (Table 3). The **equilibrium** policy has the best worst case over Red's policies (−116) and, against MIP-BR, half the war rate of MIP-BR itself (0.10 vs. 0.20). Greedy and MIP-BR do best against weaker opponents but escalate more. **Maximin** is the safest (war rate ≤ 0.08) but gives up the most control: it guards against a worst-case Red template that Red rarely plays.

MIP vs. greedy: Over 1,524 Blue decisions in 30 games the greedy heuristic matched the MIP optimum in 67% of decisions but broke the risk budget in 14%, so the MIP is needed to enforce the feasible region exactly; in play the two perform about the same.

4.5 RQ3: robustness to weather and resilient designs

As in Lecture 1, we report each Blue policy or design x's expected payoff $\sum_\omega \pi_\omega J(x,\omega)$ and worst case $\min_\omega J(x,\omega)$.



Design	calm	nom.	typh.	extr.	exp.	worst
D0 baseline	−164/.24	−144/.20	−134/.12	−119/.10	−144	−164
D1 dispersed	−153/.33	−141/.28	−131/.24	**−98**/.14	−137	−153
D2 sensing	−181/.23	−164/.23	−149/.12	−136/.11	−162	−181
D3 hotline	**−140**/.10	**−118**/.05	**−111**/.02	−102/.01	**−120**	**−140**

Figure 3: Left: Blue payoff by policy and weather. Right: designs (MIP-BR vs. MIP-BR), u_B / war rate (s.e. $\approx$ 7).

Weather: Bad weather helps Blue slightly: most policies do worst in *calm* seas (Fig. 3), since rough seas halve the effect of Red's coast-guard hulls and storms pause operations and incidents (MIP-BR war rate 0.24 calm, 0.10 extreme). The ranking is robust: MIP-BR ($\varepsilon = 0.40$) and equilibrium are first and second in all four scenarios, and the only change below them is a swap within noise. So the best policy on average is also best in the worst case. Caution: the top scorer also has the most wars (0.12–0.29), so $W = 200$ may be too small.

Designs: The hotline (D3) is best in expected and worst-case terms and cuts the war rate fourfold (0.20 → 0.05, nominal). It also raises Red's payoff (101 → 144): fewer wars help both sides. We charged it nothing, so part of this

win is by assumption; Workshop 2 will give it a cost. Dispersed basing (D1) barely changes Blue's payoff but cuts Red's ($101 \to 16$), at a higher war rate (0.28). Redundant sensing (D2) shows no measurable benefit even at zero cost: better Blue information barely changes Blue's decisions in this model.

Limitations: Parameters are assumptions; policies look one period ahead; the war rate depends strongly on h, μ and W; payoff differences below about 20 are ties (s.e. 5–12).

5 Prototype

Current state: `wargame_gui.py` runs the model of §3 as a **GUI** (Fig. 4: pick each side's plan, Blue's design and the weather, then step through turns or run many games; a Help button explains the map), as **batch** runs, or as a turn-by-turn **trace**. `experiments.py all` regenerates every number in §4; instructions are in `README.md`.

How the code reflects the model: `Game.step` is the timeline of Fig. 1 with (1)–(4); `MIPBRPolicy` solves the MIP of §3.6 each period; `MetaGame` builds Γ and runs IESDS, NE enumeration and both LPs.

Still to build: Learning agents, an attacker that reacts to Blue, Bayesian beliefs and multi-period policies (§6).

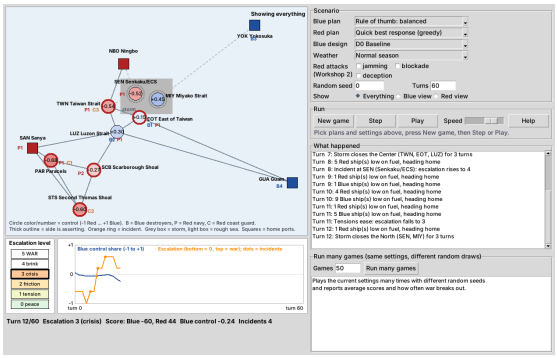


Figure 4: The GUI after 12 turns (storm in the North).

6 Plan for Workshop 2

Model: Bayesian beliefs (Lectures 7–8), multi-period best response, $\pm 50\%$ sweep on V, h, W, μ.

Learning (Module 2, once taught): A Red agent that learns from repeated play (reinforcement learning), and fictitious-play or no-regret updates on the template game. Do they converge or cycle, drift toward restraint or assertion, and exploit the predictable rule-based Blue policy?

Attacks and defenses (Module 3): Run the three coded Red attacks (CCG blockade of Blue's edges at SCB/STS, jamming of Blue sensing, false contacts) and an attacker that moves after seeing Blue's posture (backward induction). Defenses: randomized patrols, D1, D2; metric: change in control and war rate.

Resilient designs (Module 4): D0–D3 and combinations under attack and a rare long typhoon; price the hotline.
Timeline: Nov 3 beliefs, attacks; Nov 17 learning; Dec 1 redesigns; Dec 8 report.

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